

PE 810 Applied AI for Business

PSEG Weather Impact Analysis

Week 6 Final Project Report

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6-29-2026

Purpose

This report documents the design decisions completed during Week 6 of the PSEG Weather Impact Analysis project. It summarizes the research question, data sources, pipeline architecture, AI-assisted design process, and project readiness prior to implementation in Week 7.

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1. Project Overview

Focus Area

Weather Impact on Energy Systems

Research Question

How do extreme weather conditions influence electricity demand within the PJM Interconnection, and what operational insights can PSEG derive to improve energy planning?

This project develops an AI-assisted data analysis pipeline that integrates electricity demand and daily weather data to identify how weather conditions influence electricity demand. The resulting analysis provides operational insights that can help utility companies such as PSEG improve demand forecasting, resource planning, and operational decision-making.

2. Data Sources

The project uses two authoritative public data sources.

U.S. Energy Information Administration (EIA)

The EIA Open Data API was used to retrieve daily electricity information for the PJM Interconnection during calendar year 2025.

The dataset includes:

- Daily electricity demand
- Day-ahead demand forecast
- Net generation
- Total interchange

The retrieved data was exported to CSV for use within the analysis pipeline.

Source: <https://www.eia.gov/opensdata/>

National Oceanic and Atmospheric Administration (NOAA)

Daily weather observations for Newark Liberty International Airport were downloaded from NOAA for calendar year 2025.

The weather dataset includes:

- Daily precipitation (PRCP)
- Maximum temperature (TMAX)
- Minimum temperature (TMIN)

Average temperature (TAVG) will be calculated during the data cleaning process when necessary.

Source: <https://www.ncei.noaa.gov>

3. Data Access Verification

Data access has been successfully verified for both project data sources.

The EIA Open Data API was successfully queried using a registered API key, and the resulting dataset was exported to CSV format.

The NOAA weather dataset was successfully downloaded in CSV format for the selected weather station and study period.

The datasets are stored locally in the project's **data** directory as *pjm_daily_2025.csv* and *newark_weather_2025.csv*. These files will serve as the primary inputs to the analysis pipeline.

4. Pipeline Design

The project follows a modular data analysis pipeline. Each component performs a single responsibility and produces an output that serves as the input for the next stage of the workflow.

The complete pipeline design is provided in **Appendix A**.

The pipeline includes:

- Loading electricity data
- Loading weather data
- Cleaning and standardizing both datasets
- Merging datasets
- Generating analytical metrics
- Producing visualizations
- Executing the complete workflow through a single orchestration script

Each component will be developed, tested, and validated independently before being integrated into the final Python application.

Data cleaning decisions, including date standardization, numeric conversion, average temperature calculation, missing value handling, and dataset alignment, are documented within the pipeline specification.

5. AI-Assisted Development

Large Language Models (LLMs) were used to support project planning and design throughout Week 6. AI was used to review the proposed pipeline architecture and recommend improvements to component organization, modularity, and data processing strategy. Implementation will follow a component-based development approach, with each pipeline component developed, tested, and validated individually before integration into the final application. All AI interactions are documented in Appendix B. Final engineering decisions, project organization, and implementation planning remained under the control of the project author.

6. Dashboard Decision

The project will attempt to develop a simple interactive dashboard using Streamlit as a stretch goal. The dashboard will consume the outputs generated by the analysis pipeline rather than recomputing analytical results. If a fully functional dashboard cannot be completed, the development effort, AI prompts used, and lessons learned will be documented as part of the final project reflection.

7. Week 6 Status

At the conclusion of Week 6, the following milestones have been completed:

- Research question finalized
- Project data sources identified
- Data access successfully verified
- Project folder structure established
- Pipeline designed and documented
- LLM prompts log created
- Dashboard approach documented
- Component-based AI-assisted development approach documented and aligned with course requirements.

All Week 6 checkpoint requirements have been completed. The project is prepared to begin implementation and testing of the pipeline components in Week 7.

Appendices

Appendix A – Pipeline Design

Pipeline Design

Load EIA Data:

Input: *pjm_daily_2025.csv*

Output: Raw EIA DataFrame

Process: Reads "*pjm_daily_2025.csv*" from the "*data/*" folder and loads it into a Pandas DataFrame for processing.

Load NOAA Data:

Input: *newark_weather_2025.csv*

Output: Raw NOAA DataFrame

Process: Reads "*newark_weather_2025.csv*" from the "*data/*" folder and loads it into a Pandas DataFrame for processing.

Clean EIA Data:

Input: Raw EIA DataFrame

Output: Cleaned EIA DataFrame

Process: Converts the Date field to a standard datetime format, converts numeric fields to the appropriate data types, validates data quality, and prepares the dataset for analysis.

Clean NOAA Data:

Input: Raw NOAA DataFrame

Output: Cleaned NOAA DataFrame

Process: Converts the Date field to a standard datetime format, converts weather measurements to numeric values, calculates Average Temperature (TAVG), checks for missing values, and prepares the dataset for merging.

Merge Datasets:

Input:

- Cleaned EIA DataFrame
- Cleaned NOAA DataFrame

Output: *merged_data.csv*

Process: Joins both datasets on the Date field after date standardization to ensure temporal alignment between electricity demand and weather observations.

Generate Metrics:

Input: merged_data.csv

Output: Metrics tables

Process: Calculates monthly demand summaries, forecast error, weather-demand correlations, comparative statistics across seasonal and weather conditions, and at least one normalized metric to support the research question.

Create Visualizations:

Input: merged_data.csv

Output: Charts and figures

Process: Generates labeled charts and figures from the merged dataset and saves them to the output folder.

Final Orchestration:

Input: All pipeline components

Output: Final datasets, metrics, visualizations, and dashboard-ready outputs

Process: Executes the complete pipeline by integrating all components into a single workflow.

Dashboard Decision

A simple interactive dashboard will be attempted as a stretch goal. The dashboard will consume the outputs generated by the analysis pipeline (metrics and visualizations) rather than recomputing them. If a fully functional dashboard cannot be completed, the development process, prompts used, and lessons learned will be documented as part of the project reflection.

Normalized Metric

The project will calculate at least one normalized metric to support comparison of electricity demand across varying weather conditions. The specific metric will be selected based on exploratory analysis of the merged dataset and the requirements of the research question.

Appendix B – LLM Prompts Log

LLM Prompts Log

This appendix documents the LLM interactions used during the Week 6 project design phase. Each entry includes the prompt, its purpose, and how the resulting output was evaluated and incorporated into the project.

Prompt 1 – Pipeline Design Review (Required LLM Use #1)

Date: June 29, 2026

Objective

Review the proposed data analysis pipeline to ensure it satisfies the project requirements and follows good software engineering practices.

Prompt

Review the proposed pipeline for a project analyzing the relationship between weather conditions and electricity demand within the PJM Interconnection. Suggest improvements to the pipeline design, component organization, data cleaning strategy, and overall workflow.

Outcome

The recommendations were reviewed and selectively incorporated into the project. Suggestions that improved the organization, modularity, and clarity of the pipeline were adopted, while the overall pipeline design, research direction, and implementation decisions remained under the control of the project author.

Notes

The LLM was used as a design review and brainstorming tool rather than as the primary designer. Final decisions regarding the pipeline architecture, project organization, data processing strategy, and implementation approach were made by the project author after evaluating the recommendations.